Documentation

A guide to using LaTeX File Template v.3.0.0.

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Contents

1 Introduction

Thank you for choosing to use LaTeX File Template. If you find any bugs or want some feature not currently included, then my contact information can be found on my GitHub: https://github.com/MatthewEmer.

This document is intended for academics (with special focus on those in the fields of computer science or mathematics) who want to write papers or other related documents in a consistent format which ties in with standard practises from other published papers. This guide consists of setup instructions to download the files and create your first document, before moving onto the custom commands/environments designed to speed up academic writing. Before using this guide, I would suggest that you have some practise with some basic LaTeX commands using a guide such as https://www.overleaf.com/learn.

This guide is accurate as of Version 3.0.0.

2 Document Setup

2.1 Project Files

The actual LaTeX file template is contained in the folder *The Template* and consists of four files: *template.tex*, *_documentContent.tex*, *_appendix.tex*, and *template.pdf*. To use the template, all four files must be downloaded and placed in the same folder.

2.1.1 _appendix.tex

This file allows you to add an optional appendix to the end of your document which is not counted towards your page count. By default it contains a unnumbered part title which has been inserted into the table of contents.

2.1.2 _documentContent.tex

This file allows you to add all the main content of the document you are trying to write. It will be inserted after the table of contents, but before the appendix (if you choose to add one). By default it contains an example section and definition. Only pages in this file are counted towards the page count of the document.

2.1.3 template.pdf

This file contains the output of the LaTeX compiler. If you want to rename this document while still working on it, then delete this version, alongside any build files your compiler has created and change the name of *template.tex*.

2.1.4 template.tex

This file sets up the structure of the document, as well as acting as a compiler for the other two *.tex* files, ensuring they get inserted into the right sections of the document. When creating your document, you need to edit lines 12-15 of this file to create the title page, alongside the document header and footer, as described in Table 1.

Line	Variable Name	Description
12	thetitle	The document's title. This will be placed in the header.
13	thesubtitle	The document's (optional) subtitle. This will be placed in the header.
14	theauthor	The list of authors. This will be placed in the footer.
15	thedate	Replace the code if you want a static date rather than date of last compile.

Table 1: Lines to edit.

2.2 Optional Files

2.2.1 Storing Images

If you choose to include figures in your project, then you will need to create a subfolder labelled *Images* where you can place the image files for the project.

2.2.2 GitHub Repositories

If you are placing the document within a GitHub repository, you may want to copy over the *.gitignore* from this repository as it is set up to ignore any LaTeX build files.

3 Custom Commands

3.1 General Tools

3.1.1 Element Spacing

Command 3.1.1.1: *separate*

The separate command inserts vertical spacing between two elements whilst simultaneously removing the new paragraph indent from the following element. You can specify the amount of spacing as described in Table 2 and as seen in Example 3.1.1.1.

No.	Name	Description	Example Value
1	Spacing	The number of points of vertical spacing.	5

Table 2: Parameters for separate.

Example 3.1.1.1: *separate*{3} produces the separation as seen between the Command line and the description of separate.

3.1.2 Text Formatting

Command 3.1.2.1: *colouredText*

The colouredText command takes the given text and outputs it in a different colour, as chosen by the writer. This can be seen in Example 3.1.2.1 and as described in Table 3.

No.	Name	Description	Example Value
1	Colour	The colour to change the text to.	blue
2	Text	The text you want outputted in colour.	Hello World

Table 3: Parameters for colouredText.

Example 3.1.2.1: *colouredText*{*blue*}{Hello World} produces Hello World.

Command 3.1.2.2: *comment*

The comment command inserts notes into the document in red so they can be found more easily. This can be seen in Example 3.1.2.2 and as described in Table 4.

No.	Name	Description	Example Value
1	Text	The text you want outputted in colour.	Hello World

Table 4: Parameters for comment.

Example 3.1.2.2: *comment*{Hello World} produces Hello World.

3.1.3 Referencing Tables and Figures

Command 3.1.3.1: *elemRef*

The elemRef command allows you to easily reference tables and figures with clickable links. This is described in Table 5 and as seen in Example 3.1.3.1.

No.	Name	Description	Example Value
1	Label	The label of the element being linked to.	tab:Parameters for elemRef.
2	Type	The type of element being linked to.	Table

Table 5: Parameters for elemRef.

Example 3.1.3.1: *elemRef*{tab:Parameters for elemRef.}{Table} produces the reference Table 5.

3.2 Academic Environments

3.2.1 Referencing Environments

Command 3.2.1.1: *preEnvRef*

The preEnvRef command allows you to easily reference academic environments with clickable links before they are defined in the document. This is described in Table 6 and as seen in Example 3.2.1.1.

No.	Name	Description	Example Value
1	Label	The label of the environment being linked to.	ex:cmd:postEnvRef
2	Type	The type of environment being linked to.	Example
3	Counter	The name of the counter for that environment.	examplenumber

Table 6: Parameters for preEnvRef and postEnvRef.

Example 3.2.1.1: *preEnvRef*{ex:cmd:preEnvRef}{Example}{examplenumber} produces the reference Example 3.2.1.1.

Command 3.2.1.2: *postEnvRef*

The postEnvRef command allows you to easily reference academic environments with clickable links after they are defined in the document. This is described in Table 6 and as seen in Example 3.2.1.2.

Example 3.2.1.2: *postEnvRef*{ex:cmd:preEnvRef}{Example}{examplenumber} produces the reference Example 3.2.1.1.

3.3 Mathematics

3.4 Computer Science